Digital Twin - Modification Guide

What to change, where, and the minimum steps to reuse the app with a new CAD model

1. Which files do I touch?

File	Modify?	Purpose
twin-config.js	**YES - the only one**	Model, MQTT, axes (variables), camera, lighting
model/your_model.glb	REPLACE	Your exported CAD model (GLB format)
twin-shared.js	No	Shared logic: telemetry panel, MQTT values, axis animation
main.js	No	Desktop / Android viewer (reads the config)
ar-iphone.js	No	iPhone 8th Wall AR (reads the config)
index.html	No	Page layout; title and telemetry rows are generated from the config
style.css	No	Styles
gltfloader-classic.js	No	GLB loader for the iPhone AR scene

2. Folder structure

```
project/
  index.html   style.css   favicon.ico
  twin-config.js        <-- edit this
  twin-shared.js   main.js   ar-iphone.js   gltfloader-classic.js
  model/
    your_model.glb      <-- replace this
```

3. Minimum steps for a new model

1. Export the model as .glb and copy it into the `model/` folder.

2. Open twin-config.js and set `model.url` to the new file, e.g. `./model/my_robot.glb`.

3. Find the node names: set `model.debugListNodes: true`, open the page, press F12 and read the list printed in the Console. Set it back to `false` afterwards.

4. Edit the `axes` list: one line per moving part / variable (section 5 of this guide).

5. Edit the `mqtt` block: host, port, username, password and topic(s).

6. Match the payload: the JSON your PLC/robot publishes must contain one field per axis (section 6).

7. Test by publishing a value (e.g. {"PosX": 200}) and checking the part moves and the panel updates.

4. Sections inside twin-config.js

Section	What it controls	Change it when...
1. GENERAL	`appTitle` (browser tab title)	Always (new project name)
2. 3D MODEL	`url`, `scale`, `debugListNodes`	Always (url). Scale 0.001 if a model exported in mm looks huge
3. MQTT	Broker host/port, credentials, topics, optional `parsePayload`	New broker, topic or payload format
4. AXES	List of variables and the model nodes they move	**Always** - this is the core of the twin
5. MOTION	`lerp` smoothing (0.05 smooth, 1 instant)	Movement feels too slow or too jumpy
6. SCENE	Background, camera start, HDR, floor, shadow area	Model is too big/small or out of view; larger models need a larger `shadowBounds`
7. LIGHTING	Light colours, intensities and positions (incl. iPhone AR)	Model looks too bright or too dark

5. Defining axes (section 4 of the config)

Each axis is one line. Add, remove or rename lines freely; the telemetry panel adapts automatically.

```
axes: [
  // Linear axis (value in mm):
  { key: 'PosX', label: 'X Axis', node: 'Slide_X', type: 'position', axis: 'z', sign: 1 },

  // Rotary joint (value in degrees), payload field is "j1":
  { key: 'J1', label: 'Joint 1', payloadKey: 'j1', node: 'Link_1',
    type: 'rotation', axis: 'y', sign: 1 },
],
```

Field	Meaning
`key`	Unique id of the variable
`label`	Text shown in the telemetry panel
`payloadKey`	(optional) Field name in the MQTT JSON. Defaults to `key`
`node`	**Exact** name of the moving object in the GLB (case-sensitive)
`type`	`position` (linear) or `rotation` (rotary)
`axis`	Local axis of the node: `x`, `y` or `z`. If the part moves in the wrong direction or plane, try another axis
`sign`	`1` or `-1` to reverse direction
`scale`	(optional) Value to scene units. Default: position 0.001 (mm to m), rotation PI/180 (deg to rad)
`unit`	(optional) Text after the value. Default: mm or degrees

Value 0 means the pose the model had when it was exported; movement is relative to that pose.

6. MQTT payload

By default the payload is a flat JSON whose field names match each axis `payloadKey` (or `key`):

```
{ "PosX": 120, "PosY": 40, "J1": 35.5 }
```

If your payload has another shape, use `mqtt.parsePayload` in the config and return an object keyed by axis `key`:

```
// payload: {"joints":[10,20,30]}
parsePayload: (p, topic) => ({ J1: p.joints[0], J2: p.joints[1], J3: p.joints[2] })
```

7. Quick troubleshooting

Symptom	Likely cause / fix
Console: "Node ... not found"	`node` does not match the GLB name. Use `debugListNodes` and copy the exact name.
Part moves in the wrong direction	Change `sign` to -1.
Part moves in the wrong plane / rotates around the wrong axis	Change `axis` (x / y / z).
Part moves too little or too much	Adjust `scale` (or check the unit of your data).
Panel stays on "Connecting..."	Check mqtt host, port (WebSocket TLS), username and password.
Connected but nothing moves	Check topic and that payload field names match `payloadKey`.
Model huge, tiny or off-screen	Adjust `model.scale` and `scene.cameraStart`; for cut-off shadows raise `scene.shadowBounds`.
Model does not load	Check the `model.url` path. Serve the folder from a web server (not by opening index.html directly).